

NEW UNIT – Organisation and the Digestive System

Write the numbers 1 ? 10 into your book.

THE NEW UNIT



Find out which harder words in the new unit you can spell.

On a brand new page

In big writing write...

ORGANISATION
and the
DIGESTIVE SYSTEM

- 1.
- 2.
- 3.
- 4.
- 5.
- 6.
- 7.
- 8.
- 9.
- 10.

Review: Swap books with the person next to you and mark their answer with a red pen. Correct any wrong spelling.

1. Organism

2. Enzyme

3. Catalyst

4. Hydrochloric

5. Emulsification

6. Haemoglobin

7. Mesophyll

8. Coronary arteries

9. Amylase

10. Phloem

Tissues and Organs

*

Keywords: Multicellular, tissue, organ, organ system, organism

Starter: Can you make the words of three organs using the letters from below. The pictures are clues.

EXT: State the function of any of these three organs.

HIMAEBNERTAROCS



Starter: Can you make the words of three organs using the letters from below.

HIMAEBNERTAROCS

Heart



Pumps blood around the body

Brain



Help coordinate the bodies
Activities

Stomach



Important role in digestion

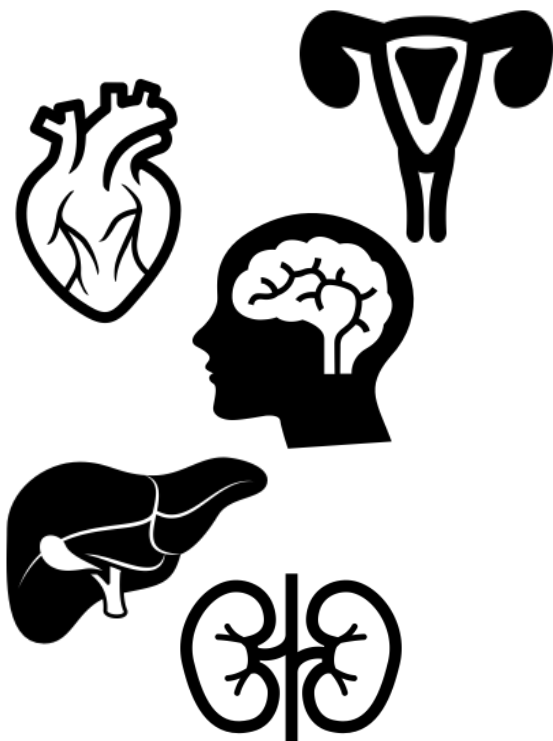
Tissues and Organs

*

Keywords: Multicellular, tissue, organ, organ system, organism, hierarchy

Starter: List as many organs in the human body as you can.
Use the pictures to help you.

EXT: Match any sentences below to any of your organs names.

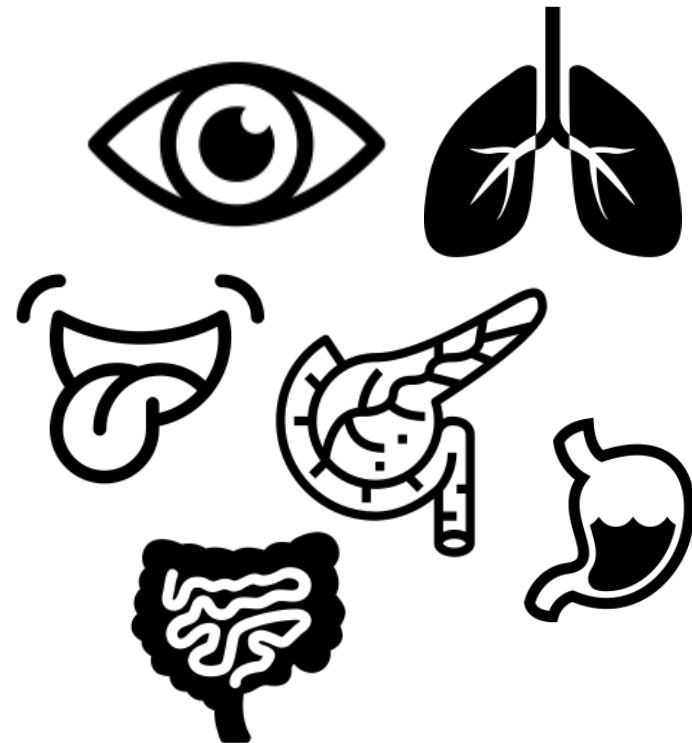


Pumps blood round
the body

Controls all the
functions of the body

Filters the blood

Where gas
exchange happens



Starter: List as many organs in the human body as you can.
Use the pictures to help you.

EXT: Match any sentences below to any of your organs names.

Ovaries



Pumps blood round
the body

Heart



Controls all the
functions of the body

Brain



Filters the blood

Liver



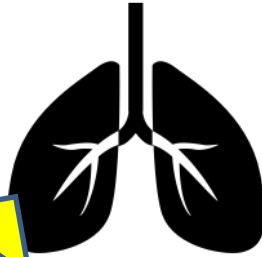
Where gas
exchange happens

Kidneys



Eyes

Lungs



Tongue



Pancrea



Stomach



Intestine

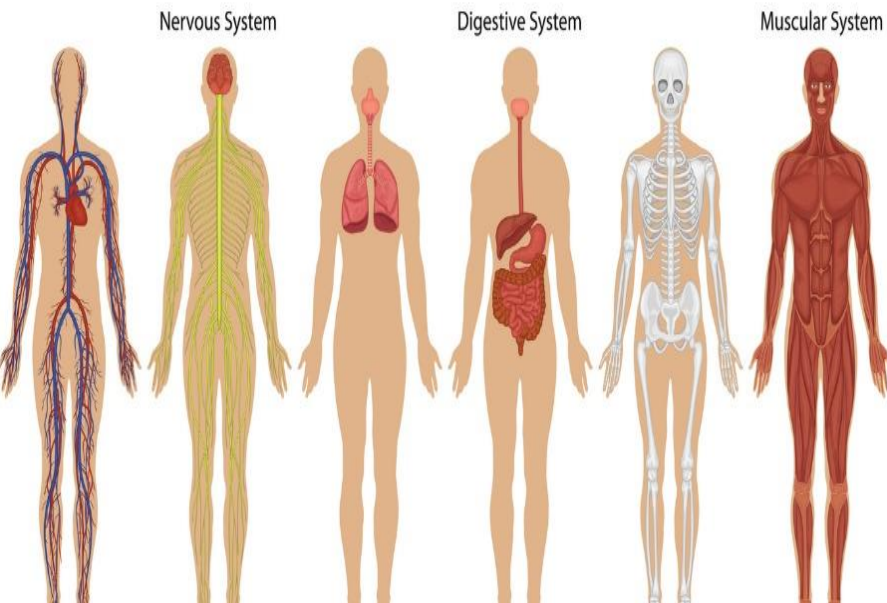


LOb: Understand the structure of our bodies.

Keywords: Multicellular, tissue, organ, organ system, organism, hierarchy

Learning Outcomes:

- State/define examples of tissues and organs and organ systems
- Explain the organisation of multicellular organisms and how they work together



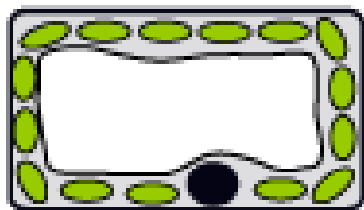
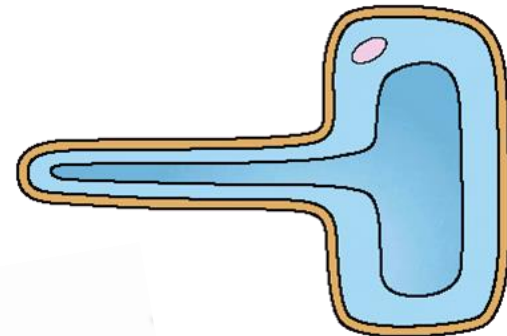
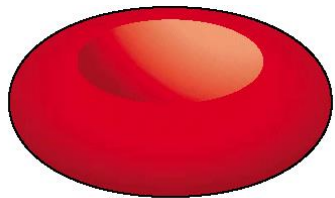
Did you know... there are 12 organ systems in our bodies.

Multicellular organisms have five layer of organisation.

Cells

Cells are the building blocks for life. They are the first level of organisation.

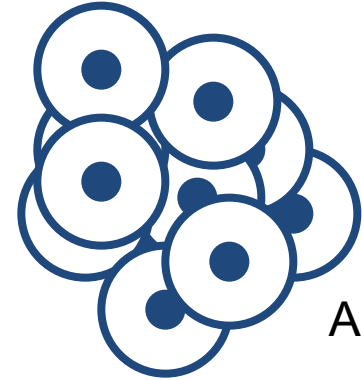
Cells, cells their made of ... organelles.



Tissues are the **second level** of organisation.



One cell



A tissue

Can you use this diagram to help come up with a definition of tissue?

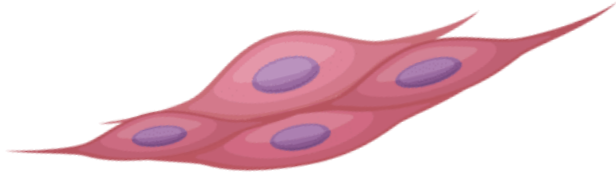
Tissue

A *group of the same type of specialised cell* working together to perform a *specific function*.

Tissues

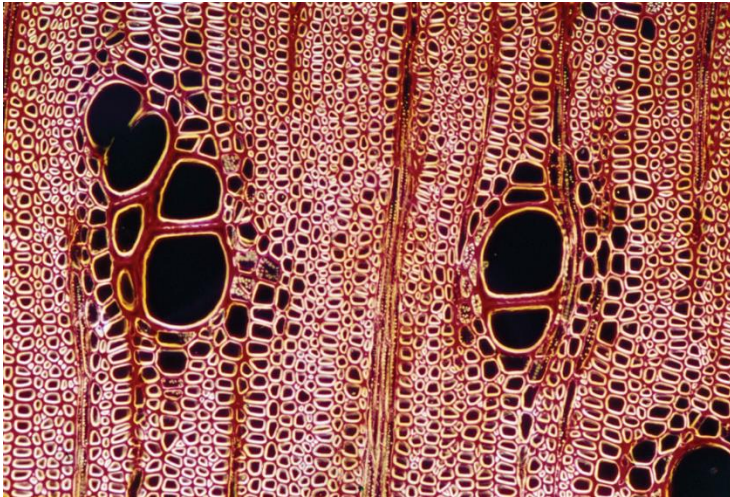
Cells

Examples

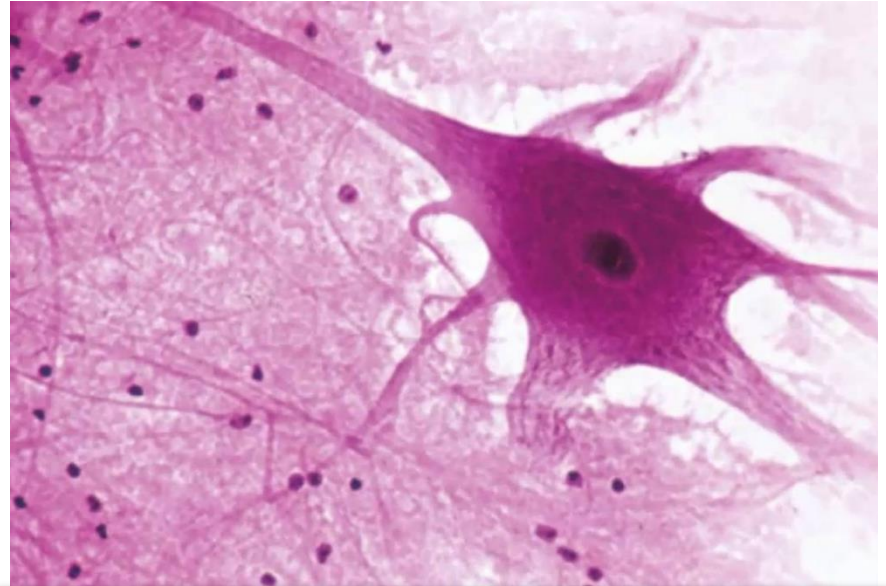


Muscle Tissue

Contract to move joints



Plant Xylem Tissue: Transport
water and minerals around the plants

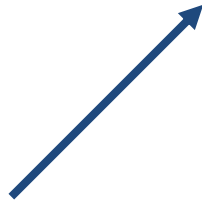
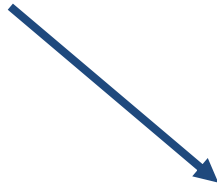
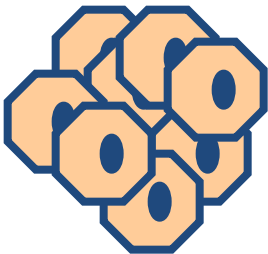
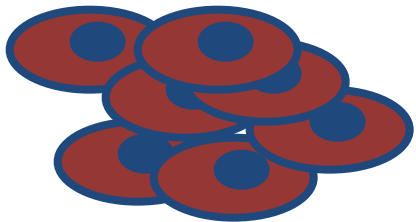
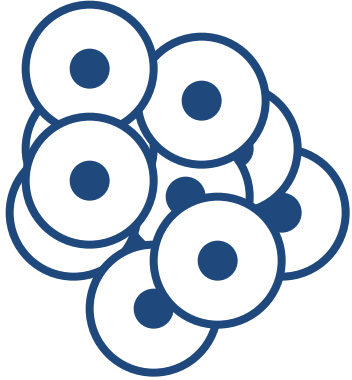


Nerve Tissue

Send electrical signals around the
body

An **organ** is a **group of different tissues** that **work together** to perform a **certain function**.

T
I
S
S
U
E



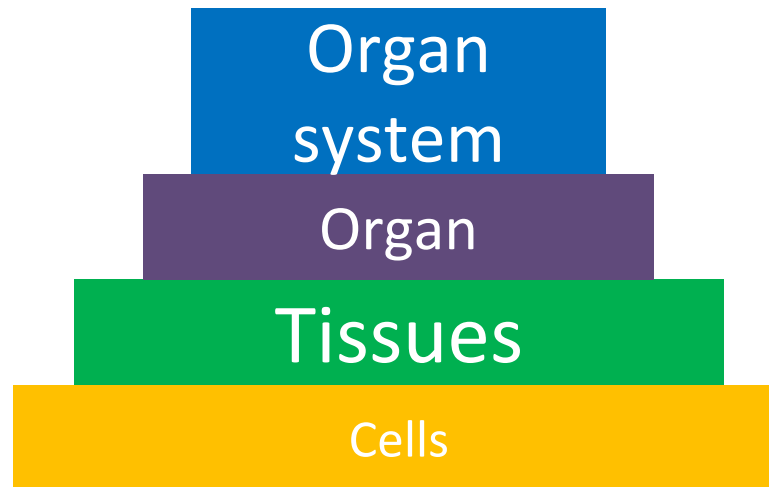
O
R
G
A
N

Organ

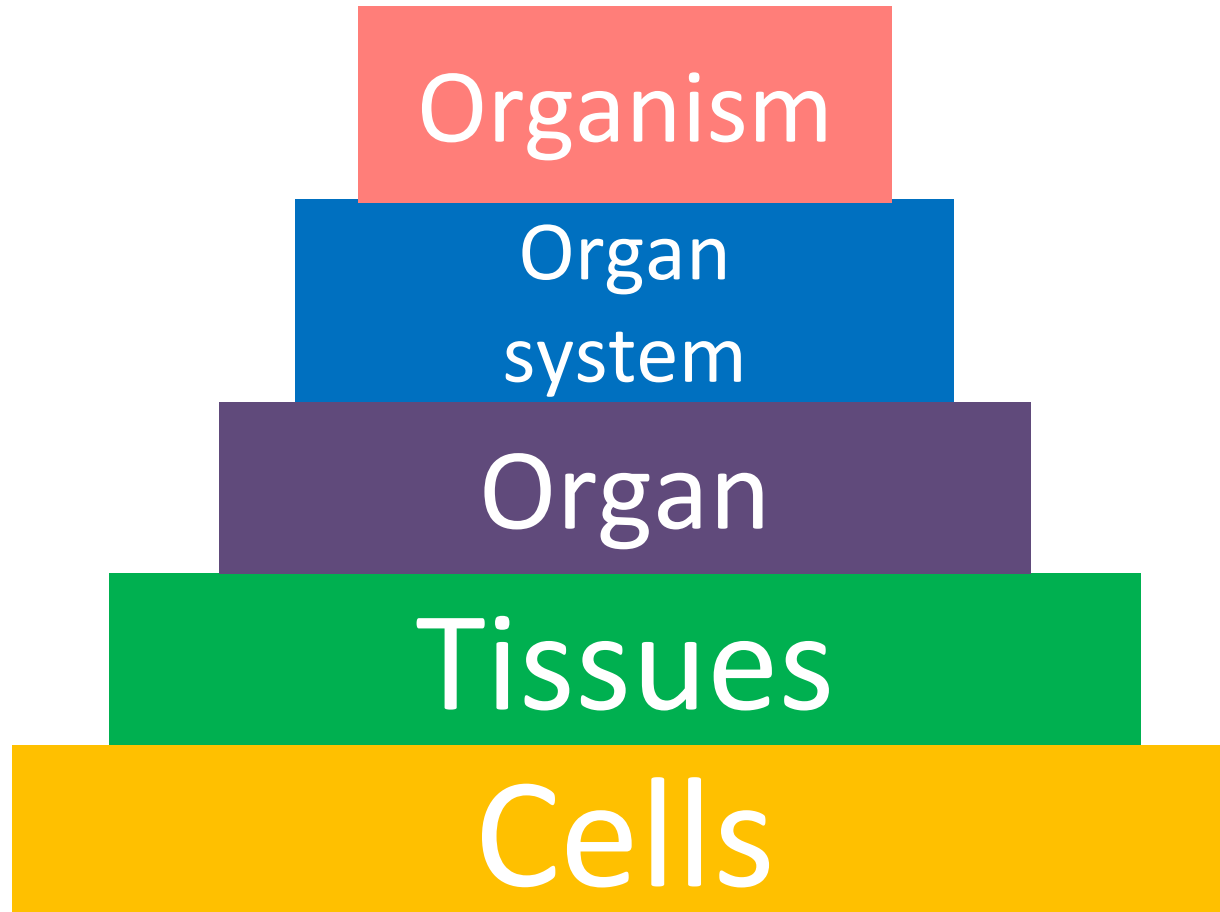
Tissues

Cells

Organ systems are the **fourth** level of organisation.



Organism is the fifth level of organisation.



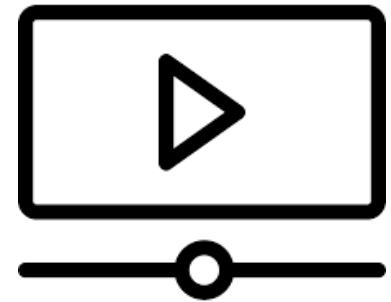
Activity: Watch the video. Write down the type of system and any organs or tissues that make up that system.

WATCH AND

LEARN

AFTERALL KNOWLEDGE IS

POWER!



Review

Muscular system - skeletal, smooth, cardiac tissues

Nervous system – Brain, spinal cord

Reproductive system -

Respiratory system – Lungs

Skeletal system – bones

Review

Circulatory system – Heart, blood vessels (arteries veins capillaries)

Digestive system – Stomach, liver, small and large intestine

Excretory system- Kidney

Organelle

Organ
System

Organism

Cell

Tissue

Organ

Human

Muscle

cell

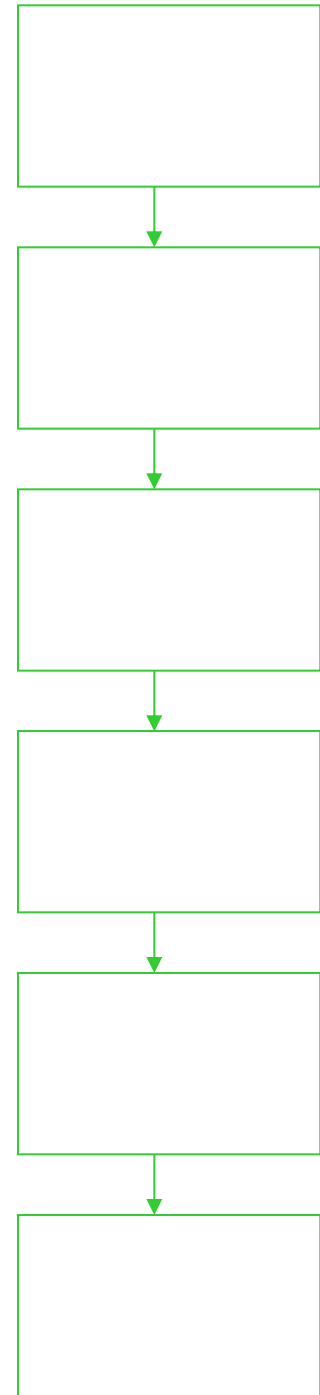
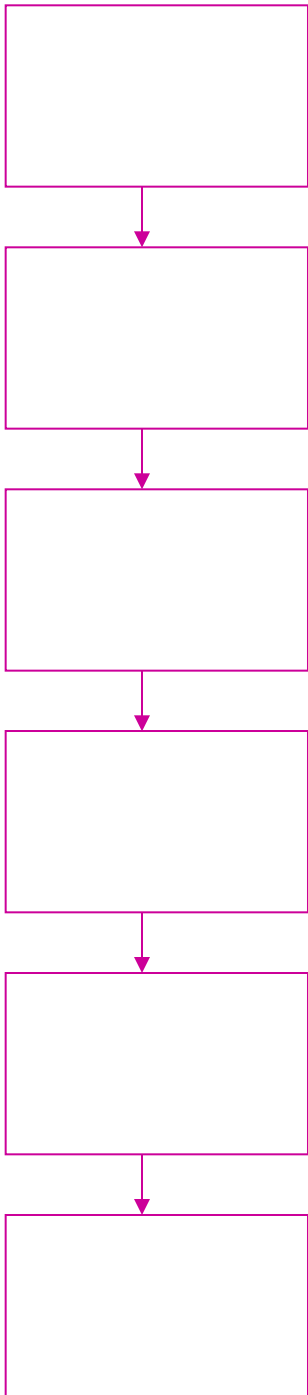
Muscle

Nucleus

Circulatory
system

Heart

As cells have lots of different functions they are often specialised to do a particular job. This means that they have special features that make them well adapted at carrying out these functions.



Activity 1: Match the keyword to the sentences

Cell

A group of similar cells working together to perform a function

Tissue

A group of different tissues working together to perform a function

Organ

A living thing

**Organ
system**

The smallest unit of any living thing

Organism

Made of different organs working together to perform a function

Review

Cell

Made of lots of the same type of cell

Tissue

Made of different cells working together

Organ

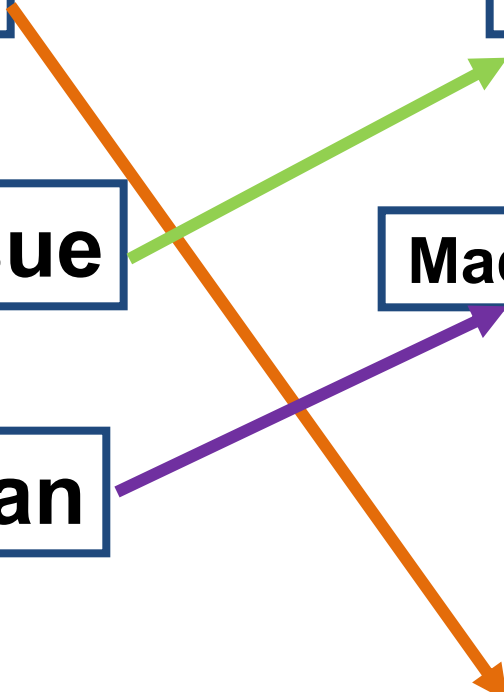
A living thing

Organ system

The smallest unit of any living thing

Organism

Made of different organs working together to perform a job



Put one tick in each row to show the level of organisation for each example.

Example	Organ	Organ system	Tissue
Stomach			
Cells lining the stomach			
Mouth, stomach, liver, intestines			
Blood			
Heart			
Muscle			
Heart, blood, blood vessels			
Brain			
Cells of the top layer of skin			

Review

Example	Organ	Organ system	Tissue
Stomach	✓		
Cells lining the stomach			✓
Mouth, stomach, liver, intestines		✓	
Blood	✓		
Heart	✓		
Muscle			✓
Heart, blood, blood vessels		✓	
Brain	✓		
Cells of the top layer of skin			✓

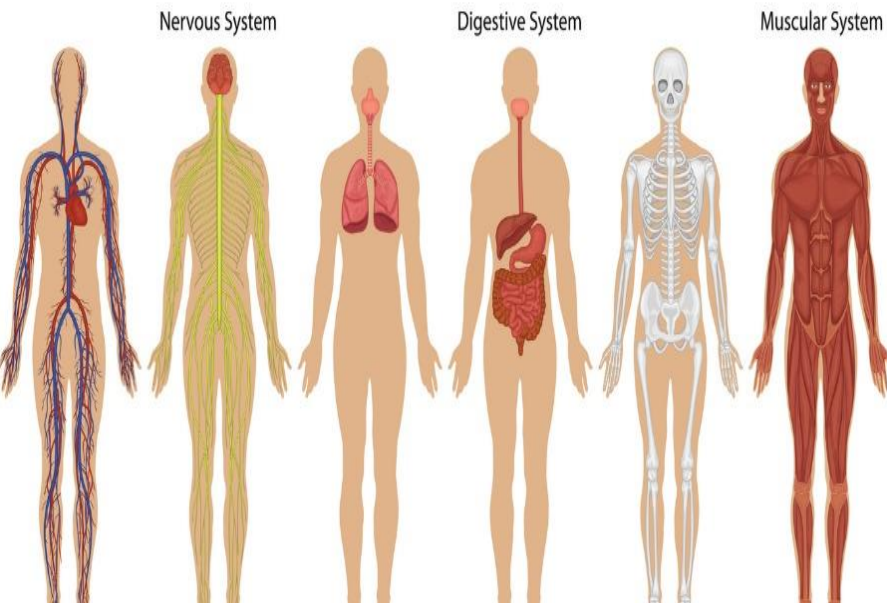
LOb: Understand the structure of our bodies.

Keywords: Multicellular, tissue, organ, organ system, organism, hierarchy

Learning Outcomes:

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➤ Explain the organisation of multicellular organisms and how they work together



Did you know... there are 12 organ systems in our bodies.

Group Task

1. **Find out** who is the oldest to youngest. The youngest is A, second youngest B
2. Next you will **read the work sheet** to familiarise yourself with it
3. I will call out a letter that person must **come to the resource area**
4. They will then go back to their group and **teach their group** so there group can fill out their work sheet
5. You can't teach your group unless you know what they need to Know so read your work sheet carefully first

EXT: Complete the other half of the first worksheet

Group Task

Circulatory System

MADE UP OF the heart, blood and blood vessels.

3 TYPES- Arteries (take blood away from the heart)
Veins (take blood to the heart)
Capillaries (tiny vessels that help supply tissues with oxygen)

One drop of blood has about **5.5 million** red blood cells and 12 thousand white blood cells.

The **heart** continues to **beat** even outside the body

Red Blood Cells Die About Every 120 Days

A human HEART is a muscle which is the size of a clenched fist.

Oxygen and nutrients are carried by this system

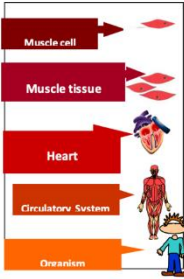


99,000km of blood vessels enough to go around the earth 2 1/2 times



If the heart stops working properly then that can cause a heart attack

Blocked vessels can be replaced using vessels from a person's own body



Respiratory (breathing) System



BloOd VesseL carrying BLOOd tAKes OxyGen FROM thE LungS to thE PaRtS of THE boDy thAT neEd iT and CARBOn dioxIde is Taken TO thE luNGS to be ReLEASED.

ORGANS

Mouth/Nose

Oesophagus

Lungs

12 to 20 BREATHS PER MINUTE.

MUSCULAR System

THREE TYPES OF MUSCLE:

SMOOTH
CARDIAC
SKELETAL

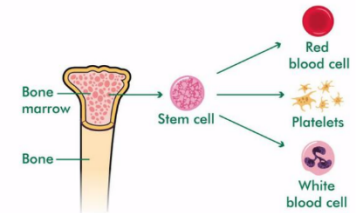


THE CIRCULATORY SYSTEM WORKS WITH THE MUSCULAR SYSTEM AS IT SUPPLIES MUSCLES WITH THE OXYGEN AND GLUCOSE IT NEEDS TO BE ABLE TO WORK VIA **arteries, capillaries and veins**

Skeletal System



206 Bones for an adult



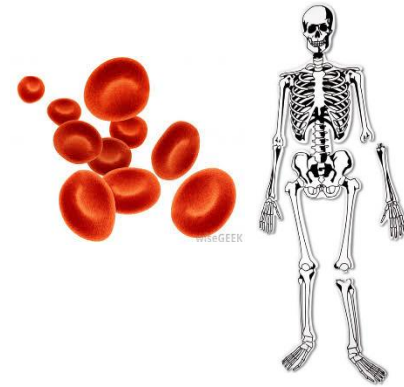
Inside some **bones** is **marrow** which makes our blood cells.

Without the skeletal system there would be no blood cells to carry oxygen for the circulatory system.

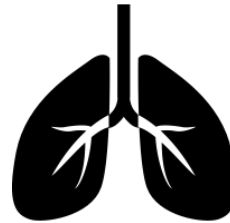
Review

How does the circulatory system work together with the other systems?

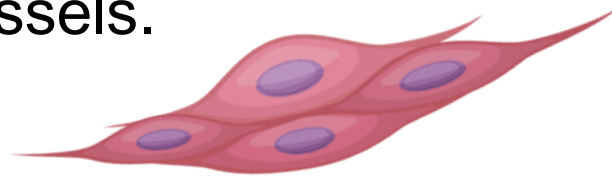
Skeletal system supplies the blood cells from bone marrow.



Blood is transported to breathing system using vessels. Oxygen is picked up and carbon dioxide is released.



Muscular system needs oxygen and glucose which is supplied To it by blood and vessels.



Review

TASK 3: Answer the questions below

Questions

1 Put the structure of the human body in order biggest to smallest. :

Organism, **ORGAN SYSTEM, ORGAN, TISSUE, CELL**

2 Complete the following Table with an appropriate organ from the following:

lung

kidney

arteries

Bronchi

liver



Organ system	Examples of organs		
excretory system	LIVER	bladder	KIDNEY
circulatory system	ARTERIES	veins	heart
respiratory system	trachea	BRONCHI	LUNG

3 Link the following organs with their function:

Pancreas	Controls the body
Kidneys	Carries food from the mouth to the stomach
Oesophagus	Produces insulin used to control blood sugar levels
Brain	Involved in excretion and the production of urine

Review

EXT

1 a Define what is meant by a tissue.

A group of the same type of specialised cell working together to perform a specific function. [1]

(1 mark)

b State **two** examples of tissues and explain their function.

Muscle Tissue: Contract to move joints [2]

Nerve tissue: Send electrical signals around the body [2]

Xylem tissues: Transports water and minerals throughout the plant [2]

(4 marks)

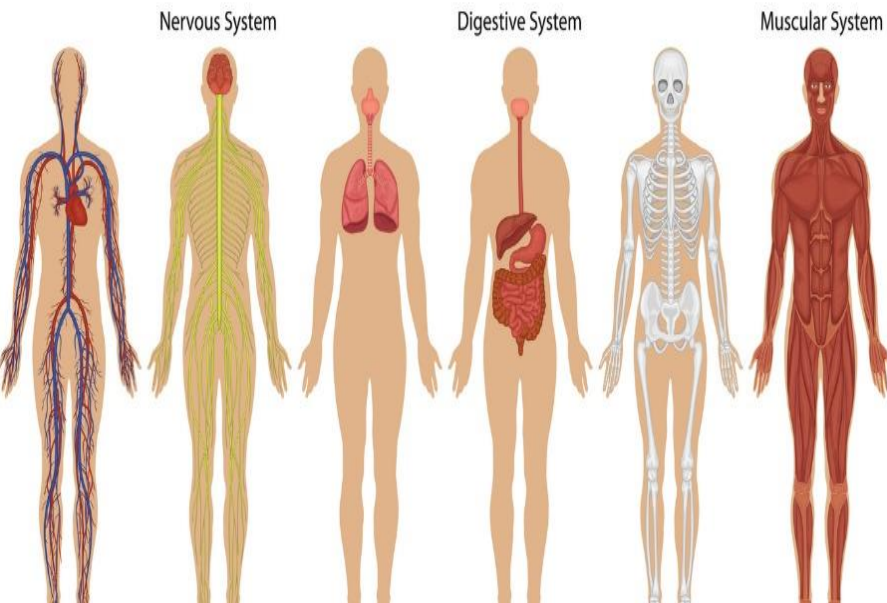
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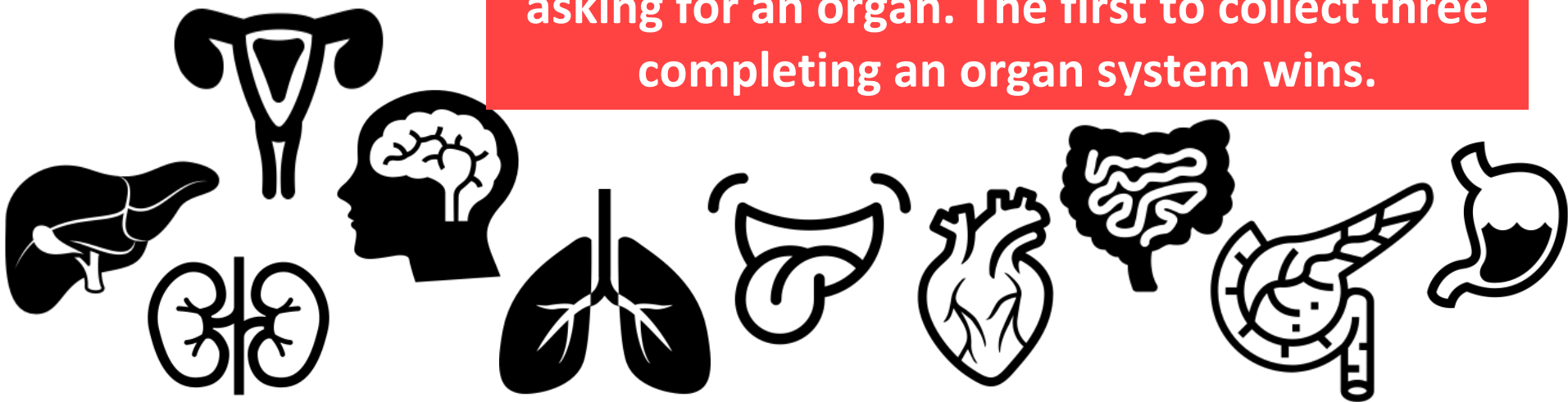


Did you know... there are 12 organ systems in our bodies.

Organ System collection game.

RULES

1. Each group will receive a pack of cards with organs in them.
2. Share the organs equally between the members.
3. First person asks any other random person for an organ. If that person has the organ they must give it to the person asking for it.
4. Each person round the circle continue asking for an organ. The first to collect three completing an organ system wins.



Organ Systems and Organs

nervous system – brain, eyes, spinal cord

digestive system – stomach, pancreas, oesophagus

excretory system – kidney, bladder, liver

respiratory system – lung, trachea, bronchi

female reproductive system – ovary, uterus, vagina

male reproductive system – penis, testes, scrotum

circulatory system – heart, arteries, veins

keyfacts



There are **5 levels of organisation**. Cells, tissues organs, organ system and organism.



Tissues are made of **similar cells** working together to perform a **certain function**.



Organs are groups of **different tissues** working together to perform a **certain function**.



Circulatory system works with the breathing system, skeletal system and muscular system.

Plenary: Fingers

Draw around your hand.

Answer the statement in each finger



Plenary: Chose a box

Write the sentence starter and complete the sentences about today's lesson.

6. The part of the lesson I found the easiest to understand was ...

8. After today's lesson, I can now explain ...